

PINGTUNG NORTH, Taiwan

WMO: 467580

Lat: 22.700N

Lon: 120.482E

Elev: 30

StdP: 100.97

Time Zone: 8.00 (E08)

Period: 86-99

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	11.1	12.7	8.8	7.1	13.7	10.0	7.7	14.5	7.1	24.3	6.1	23.4	0.4	10	0.289

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.0	34.4	27.3	34.0	27.2	33.2	26.9	28.2	32.8	27.8	32.3	27.4	31.8	4.4	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.1	22.9	30.6	26.7	22.3	30.3	26.2	21.7	29.9	91.2	32.7	89.0	32.8	87.4	31.8	30.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
7.4	6.1	5.2	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
			DB												
			WB	3.8	29.6	3.4	0.5	1.3	30.0	-0.7	30.3	-2.6	30.5	-5.1	30.9

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	25.1	19.5	20.3	22.6	25.5	27.6	28.5	29.0	28.8	28.2	26.6	23.9	20.5
	DBStd	3.97	2.36	2.56	2.92	2.22	1.43	1.55	1.36	1.40	1.25	1.46	2.02	2.41
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	40	16	11	5	1	0	0	0	0	0	0	1	8
	CDD10.0	5513	294	287	392	464	545	556	590	582	547	514	416	327
	CDD18.3	2511	52	64	138	214	287	306	332	323	297	256	166	76
	CDH23.3	24290	293	401	997	1802	2937	3491	3896	3621	3084	2264	1086	420
	CDH26.7	9077	36	78	309	643	1151	1464	1698	1468	1152	752	267	58
Wind	WSAvg	1.4	1.2	1.3	1.3	1.4	1.5	1.8	1.9	1.9	1.5	1.2	1.0	1.0
Precipitation	PrecAvg	1798	15	21	22	68	154	396	372	413	229	62	28	19
	PrecMax	2753	56	93	99	241	631	984	803	1200	785	278	202	154
	PrecMin	951	0	0	0	0	2	33	4	106	53	0	0	0
	PrecStd	492	16	25	30	72	126	273	214	298	176	75	45	33
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	29.2	30.1	32.2	33.4	34.2	35.2	35.2	34.9	34.1	33.1	31.2	29.8
		MCWB	22.4	23.1	24.8	25.6	26.8	27.5	27.3	27.5	27.3	26.2	25.0	22.9
	2%	DB	27.8	28.9	31.1	32.2	33.3	34.2	34.4	34.0	33.2	32.0	30.3	28.1
		MCWB	21.2	22.2	24.4	25.2	26.4	27.3	27.3	27.5	27.0	25.9	24.4	22.1
	5%	DB	26.3	27.7	29.9	31.5	32.8	33.7	33.9	33.2	32.7	31.2	29.2	27.0
		MCWB	20.2	21.6	23.7	25.0	26.2	27.2	27.2	27.2	26.8	25.6	23.8	21.3
	10%	DB	25.0	26.0	28.6	30.3	31.8	32.8	33.1	32.6	31.9	30.4	28.2	25.9
		MCWB	19.6	20.6	23.1	24.5	25.9	26.9	27.0	27.0	26.5	25.2	23.2	20.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	23.3	24.1	25.8	26.9	27.9	28.8	28.7	28.6	28.3	27.1	26.0	23.8
		MCDB	27.9	28.7	31.1	31.8	32.4	33.4	33.4	33.1	32.6	31.4	30.1	28.6
	2%	WB	21.9	22.8	24.7	26.0	27.2	28.1	28.1	28.0	27.6	26.5	25.2	22.7
		MCDB	26.5	27.4	30.0	30.9	31.6	32.6	32.9	32.5	31.9	30.7	29.3	27.0
	5%	WB	21.0	21.9	24.0	25.4	26.7	27.5	27.7	27.5	27.2	26.1	24.4	21.9
		MCDB	25.0	26.8	29.0	30.2	31.0	32.0	32.3	31.8	31.3	30.0	28.4	26.0
	10%	WB	20.1	20.9	23.3	24.8	26.3	27.0	27.2	27.1	26.7	25.6	23.7	21.1
		MCDB	23.8	25.4	27.8	29.4	30.5	31.5	31.7	31.2	30.5	29.3	27.2	24.8
Mean Daily Temperature Range	5% DB	MDBR	10.7	10.7	10.2	8.9	8.3	7.8	8.0	7.5	7.6	8.3	9.2	10.6
		MCDBR	12.1	12.8	12.0	10.2	9.6	9.2	9.3	8.6	8.4	8.9	9.8	11.1
		MCWBR	6.3	7.0	6.2	4.8	4.3	4.0	3.8	3.7	3.6	3.8	4.9	5.8
	5% WB	MCDBR	10.9	11.8	11.2	9.5	8.6	8.4	8.3	8.0	7.9	8.4	9.3	10.5
		MCWBR	6.4	6.8	6.2	4.8	4.2	4.1	3.9	3.8	3.7	4.0	4.8	5.7
Clear-Sky Solar Irradiance	taub	0.504	0.542	0.666	0.652	0.541	0.500	0.501	0.559	0.595	0.597	0.516	0.523	
	taud	2.014	1.940	1.680	1.754	2.077	2.215	2.204	2.013	1.890	1.844	2.048	1.977	
	Ebn at Noon	749	751	678	699	777	802	802	758	722	694	733	712	
	Edh at Noon	159	182	246	232	167	144	146	177	197	198	153	160	
All-Sky Solar Radiation	RadAvg	3.50	4.08	4.43	4.80	5.05	5.31	5.62	5.00	4.89	4.52	3.70	3.31	
	RadStd	0.33	0.40	0.38	0.48	0.56	0.60	0.47	0.53	0.42	0.26	0.26	0.33	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (21 neighbors)	+0.24	N/A	N/A	N/A	+0.20	+0.24	N/A	N/A	+116	+110

Nomenclature: See separate page